Statistical Decision Making
45-752

Broadening the Domain of Application of Regression Models

This file is named: 07_slides.pptx
Commands are in: 07_notebook.ipynb

1

Table of Contents

- ❑Choosing functional form: Choosing Functional Form
- ❑Functional form and AI
- ❑Building a Regression Model
- ❑Summary

2

Statistical Decision Making
45-752

Choosing Functional Form
Application: Building a Regression Model to Predict
Prices of Condominiums in Boston

3

Boston Condominiums Data Description

We will develop a model to predict prices of condominiums in Boston. Our variables are as follows:

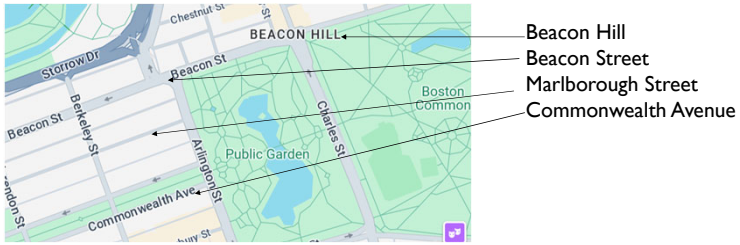
Dataset Boston_Condos_572 contains information about condominium prices and characteristics of 572 condominiums for an area of central Boston that is much sought after. The data include:

- sale price in dollars (PRICE),
- floor area in square feet (SQFT),
- number of bedrooms (BED),
- height of building in number of floors (FLOORS),
- distance from a large public park called the Boston Common (CDIST),
- an indicator variable denoting the availability of parking in the building (PARK),
- and indicator variables denoting the street on which the unit is located:
- MARST if on Marlborough Street,
- BECST if on Beacon Street,
- COMAVE if on Commonwealth Avenue, and
- BHILL if on Beacon Hill.

The data set is from some years ago. Boston condominium prices have gone up in the interim, but that does not reduce the learning value of the data set. Indeed, I think you will find this a very interesting application.

4

Map Showing Boston Streets in Our Data



5

5

Summary Statistics

To view summary statistics, I issued:

```
Boston_Condos_572.describe().transpose()
```

The mean price is $141 thousand dollars. The standard deviation is $86 thousand.

Square feet ranged from 216 to 2511. Number of bedrooms ranged from 0 to 4.

Distance to Boston Common is in thousands of feet. Distance ranged from 100 feet to roughly 10,400 feet.

The Boston Common was the nation's first public park.

	count	mean	std	min	25%	50%	75%	max
Observation	572.0	286.500000	165.266452	1.000000	143.750000	286.500000	429.25	572.0
PRICE	572.0	141247.284965	85706.850830	26000.000000	80000.000000	120750.000000	180000.00	595000.0
BECST	572.0	0.473776	0.499749	0.000000	0.000000	0.000000	1.00	1.0
BHILL	572.0	0.017483	0.131175	0.000000	0.000000	0.000000	0.00	1.0
COMAVE	572.0	0.353147	0.478366	0.000000	0.000000	0.000000	1.00	1.0
MARST	572.0	0.155594	0.362788	0.000000	0.000000	0.000000	0.00	1.0
BATH	572.0	1.417832	0.575535	1.000000	1.000000	1.000000	2.00	4.0
BED	572.0	1.431818	0.807518	0.000000	1.000000	1.000000	2.00	4.0
PARK	572.0	0.250000	0.433392	0.000000	0.000000	0.000000	0.25	1.0
SQFT	572.0	920.351399	438.958248	216.000000	588.750000	790.000000	1169.50	2511.0
FLOORS	572.0	7.603584	4.382623	3.315441	3.868594	5.365798	11.45	15.5
CDIST	572.0	5.183252	3.096217	0.100000	2.455000	5.600000	8.00	10.4

6

Estimate A Linear Regression

We wish to develop a model of price as a function of location and characteristics of the condominiums.

- Suppose we start with a linear model:

```
# Estimate Linear Regression
Reg_Condos=tprstats.model("cs", "PRICE~BATH+BECST+BED+BHILL+CDIST+FLOORS+MARST+PARK+SQFT", Boston_Condos_572)
Reg_Condos.summary()
```

Let's conduct a Ramsey test of the functional form:

```
# Test functional form. We obtain low p values
Reg_Condos.ramsey_test()
```

The Ramsey test yields low p-values, suggesting this may not be a good functional form.

power	pvalue
2	2.398720e-02
3	1.594274e-10

7

7

Thinking About The Model

If we think about the application, it becomes clear that the linear specification is not very plausible for this application.

For example, the linear specification assumes that the value of an additional bathroom is the same regardless of the size of the condominium or the number of bedrooms. This seems unlikely.

Also, we would expect that prices of condominiums would be proportional to size (SQFT) rather than linear in size.

- Hence, it is natural to think of taking logarithms.

The histograms next slide show that price is quite skewed while the logarithm of price is relatively symmetric. We see the same for square feet of floor space.

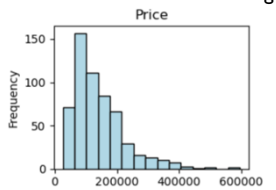
- When a variable is right skewed and the log of the variable is relatively symmetric, it is often the case that the logarithm of the variable is the better choice.

8

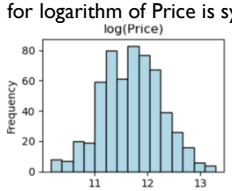
8

Histograms of Price and Log Price

Histogram for Price is skewed to the right:



Histogram for logarithm of Price is symmetric:

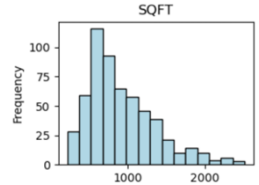


9

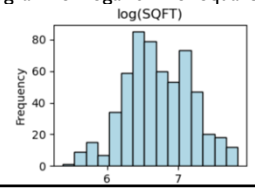
9

Histograms of Square Feet and Log Square Feet

Histogram for Square Feet is skewed to the right:



Histogram for logarithm of Square Feet is relatively symmetric:



10

10

Regression with logarithms

In the following regression, PRICE and SQFT are in log form:

```
# Investigate functional form
Log_Reg_Condos1=tprstats.model("cs", "log(PRICE)~log(SQFT)+BATH+BED+CDIST+FLOORS+PARK+BECST+BHILL+MARST", Boston_Condos_572)
Log_Reg_Condos1.summary()
```

- The result is on the next slide.

Before viewing the estimates, let's conduct the Ramsey test.

```
# Ramsey test
Log_Reg_Condos1.ramsey_test()
```

We obtain much higher p-values than in the previous linear specification. This suggests we have an improved functional form.

power	pvalue
2	0.166757
3	0.374514

11

11

PRICE and SQFT in log form

The regression has a coefficient of 1.00 for log(SQFT). This is very intuitive. It tells us that, holding other variables constant, a one percent increase in square feet increases predicted price by 1%.

The coefficient of CDIST is insignificant. I found this somewhat surprising.

I tried the logarithm of CDIST.
See next slide.

OLS Regression Results

Dep. Variable:	log(PRICE)	**R-squared:**	0.831
Model:	OLS	**Adj. R-squared:**	0.828
No. Observations:	572	**F-statistic:**	305.7
Covariance Type:	HC1	**Prob (F-statistic):**	5.07e-210

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	4.8367	0.290	16.676	0.000	4.267	5.406
log(SQFT)	1.0045	0.048	20.864	0.000	0.910	1.099
BATH	0.0335	0.031	1.094	0.274	-0.027	0.094
BED	0.0281	0.029	0.970	0.332	-0.029	0.085
CDIST	0.0046	0.004	1.212	0.226	-0.003	0.012
FLOORS	-0.0102	0.003	-4.019	0.000	-0.015	-0.005
PARK	0.1406	0.028	5.041	0.000	0.086	0.195
BECST	0.0597	0.024	2.477	0.014	0.012	0.107
BHILL	0.3012	0.060	5.013	0.000	0.183	0.419
MARST	0.0552	0.034	1.603	0.109	-0.012	0.123

12

12

Boston Condos Model

We see that the coefficient of log(CDIST) is significant at the 5% level.

- Interpret the coefficient of log(CDIST). Hint: The coefficient is an elasticity.

Which other coefficients are significant at the 5% level?

What is the reference category for streets?
Hint: The streets are:
-Commonwealth (COMAVE)
-Beacon Street (BECST)
-Beacon Hill (BHILL)
-Marlborough street (MARST)

Dep. Variable:	log(PRICE)	R-squared:	0.832
Model:	OLS	Adj. R-squared:	0.830
No. Observations:	572	F-statistic:	321.7
Covariance Type:	HC1	Prob (F-statistic):	3.51e-215

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	4.8847	0.286	17.055	0.000	4.322	5.447
log(SQFT)	0.9914	0.048	20.677	0.000	0.897	1.086
BATH	0.0370	0.030	1.217	0.224	-0.023	0.097
BED	0.0306	0.029	1.056	0.291	-0.026	0.088
log(CDIST)	0.0283	0.011	2.483	0.013	0.006	0.051
FLOORS	-0.0084	0.003	-3.021	0.003	-0.014	-0.003
PARK	0.1324	0.028	4.776	0.000	0.078	0.187
BECST	0.0699	0.025	2.789	0.005	0.021	0.119
BHILL	0.3442	0.058	5.954	0.000	0.231	0.458
MARST	0.0637	0.035	1.821	0.069	-0.005	0.132

13

13

Interpretation of Coefficients on Previous Slide

Practice interpreting semi-log coefficients:

Interpret coefficient of BECST.

- Other things constant, a unit on Beacon Street sells for 7% more than a unit on Commonwealth Avenue.

Interpret coefficient of FLOORS. Hint: The coefficient is less that .15 rule of thumb.

What is the effect on price of having a parking space in the building?

Interpret coefficient of BHILL.

- 100*(exp(.344*1)-1) We obtain 41.06%
- Other things constant, a unit on Beacon Hill sells for 41% more than a unit on Commonwealth Avenue.

Interpret the coefficient of BED.

Practice interpreting log-log coefficients (hint elasticity):

- Interpret coefficient of log(SQFT).
- Interpret coefficient of log(CDIST).

14

14

Hypothesis Tests

Let's test the functional form on the previous slide using the Ramsey Test.

High p-values are favorable to this functional form.

power	pvalue
2	0.241611
3	0.499855

Test the null hypothesis that the logarithm of SQFT equals 1.

```
# Let's test the null hypothesis that the coefficient of log(SQFT) equals 1
Log_Reg_Condos2.wald_test("log(SQFT)=1")
```

We obtain p-value .858. Hence, we do not reject.

Test the null hypothesis that the coefficient of BATH equals the coefficient of BED

```
# Test the hypothesis that coefficint of BED equals coefficient of BATH
Log_Reg_Condos2.wald_test("BED=BATH")
```

We obtain p-value .893. Do we reject or not reject?

15

15

Investigate Some Pairwise Correlations

In the regression, we see that BED and BATH have high p-values.

The following correlations give insight into why.

```
# Calculate correlations
Boston_Condos_572["BED"].corr(Boston_Condos_572["BATH"])
```

0.636

```
Boston_Condos_572['BED'].corr(Boston_Condos_572['log_SQFT'])
```

0.807

```
Boston_Condos_572['BATH'].corr(Boston_Condos_572['log_SQFT'])
```

0.723

Given the high correlation of SQFT with both BED and BATH, we might expect that log(SQFT) would also be insignificant.

However, there is much more variation in log(SQFT) than in BED and BATH as shown on the histograms on the next slide.

16

16

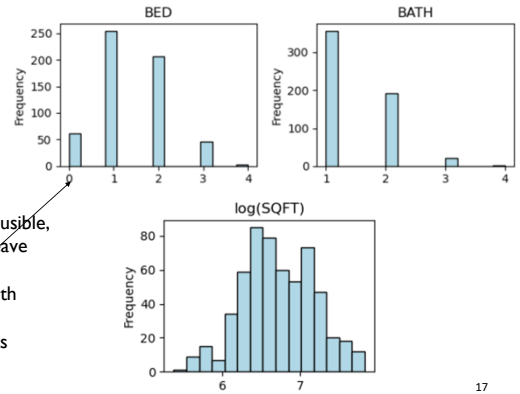
Histograms of Log(SQFT), BED, BATH

These histograms give insight into why the logarithm of SQFT is more significant than the BED and BATH variables.

There is more variation in log(SQFT) than in the BED and BATH variables.

While BED and BATH have relatively high p-values, their coefficients are plausible, and we certainly would not want to leave those variables out of the model.

- It would be hard to sell a condo with no bathrooms.
- Interestingly, about 10% of the units have no bedrooms.



17

17

Multicollinearity

Our Boston Condominimum application illustrates why variables that are relatively highly correlated with each other can have insignificant coefficients.

Think of an extreme case. Suppose two explanatory variables were perfectly correlated. It would then be impossible to estimate the separate effects of the two variables.

If two explanatory variables are highly correlated but not perfectly correlated, we can obtain estimates of their coefficients. However, we may not get very precise estimates of their individual effects if they tend to move fairly closely together. Hence, we will obtain relatively high p-values for the coefficients.

This is an example of what is called multicollinearity.

To help you understand multicollinearity in this application, the next slide illustrates the effect of leaving out a variable that should be in the model.

18

18

Illustration of Effect of Multicollinearity

Suppose we leave out the square foot variable:

```
Log_Reg_Condos3=tprstats.model("cs", "log(PRICE) ~ BATH + BED + log(CDIST) + FLOORS + PARK + BECST + BHILL + MARST", Boston_Condos_572)
Log_Reg_Condos3.summary()
```

Note the nearly 10 fold increase in the coefficients of BED and BATH, and note their exceedingly low p-values.

Of course, it would be foolish to leave out the square foot variable. I did so here to demonstrate that leaving out a variable that belongs in a model can result in large changes in coefficients of variables correlated with the left out variable.

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	10.6754	0.069	154.852	0.000	10.540	10.811
BATH	0.3114	0.040	7.853	0.000	0.234	0.389
BED	0.3575	0.029	12.261	0.000	0.300	0.415
log(CDIST)	0.0732	0.017	4.253	0.000	0.039	0.107
FLOORS	-0.0123	0.004	-3.029	0.003	-0.020	-0.004
PARK	0.1780	0.038	4.680	0.000	0.103	0.253
BECST	0.0343	0.034	1.001	0.317	-0.033	0.102
BHILL	0.4565	0.093	4.934	0.000	0.275	0.638
MARST	-0.0184	0.048	-0.382	0.703	-0.113	0.076

19

19

Prediction

Suppose the following condominium has come on the market. The unit is on Beacon Street with 2 bedrooms, 2 bathrooms, parking in the building, CDIST=6.300, 4 floors in the building, and 1000 square feet of space. Predict the price at which it will sell and form a 75% prediction interval.

```
Xnew = pandas.DataFrame({
    "SQFT": [1000], "BATH": [2], "BED": [2], "CDIST": [6.3], "FLOORS": [4],  "PARK": [1], "BECST": [1],  "BHILL": [0],"MARST": [0]
    })
Log_Reg_Condos2.prediction_intervals(Xnew, alpha=0.25).map(numpy.exp)
```

Predicted	Lower	Upper
177929.831399	135174.303659	234208.899507

- The predicted price is $177,930
- The 75% prediction interval is [$135,174, $234,209]

This data set is from many years ago, so prices are much higher in Boston now. However, this is an excellent data set from a learning perspective.

20

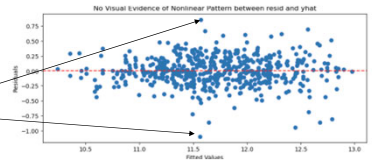
20

Plot Residuals Against Fitted Values

The following commands calculate residuals and fitted values and plot former vs latter with a red horizontal line at zero.

```
plt.figure(figsize=(10, 4))
plt.scatter(yhat, resid,color='b')
plt.axhline(y=0, color='r', linestyle='--')
plt.scatter(yhat, resid)
plt.xlabel('Fitted Values')
plt.ylabel('Residuals')
plt.title('No Visual Evidence of Nonlinear Pattern between resid and yhat')
plt.show()
```

We see some evidence of heteroscedasticity, (more variability toward the mid range of yhat). This was automatically addressed when we estimated the model.
There are two relatively large outliers. Let's investigate on next slide.



21

21

Investigating the outliers

Find the outliers:

```
# Find and view the index of the large negative residual
index1=resid.argmax()
index2= resid.argmin()
index1,index2
```

```
(np.int64(5), np.int64(368))
```

We see above that Observations 6 and 369 have the largest and smallest residuals.

Let's view the data for those two condos. `Boston_Condos_572.iloc[['5', '368']]`

Comparing to the summary statistics presented earlier, we see that Observation 6 has below median floor area, 1 bathroom and no bedrooms.

By comparison, Observation 369 has 1 bathroom, 2 bedrooms and larger floor area. Perhaps there are distinctive features not measured that make the first condo highly desirable and the second not very desirable.

	Observation	PRICE	BECST	BHILL	COMAVE	MARST	BATH	BED	PARK	SQFT	FLOORS	CDIST	STREET	log_PRICE	log_SQFT	log_CDIST
5	6	250000	0	0	0	1	1	0	1	668	3.417500	3.34	marst	12.429216	6.504288	1.205971
368	369	35000	0	0	1	0	1	2	0	775	7.403409	5.00	comave	10.463103	6.652863	1.609438

22

Statistical Decision Making
45-752

Functional Form and AI

23

Data Set Information

In the data sets for the course, you will find a file named: Functional_Form_Data_AI.xlsx

This is a data set that I made up. The variables in the file are named Y, X1, X2

We now investigate the data and choose a functional form.

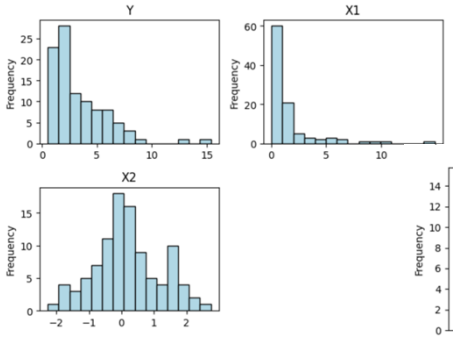
Then we will ask AI to choose a functional form.

24

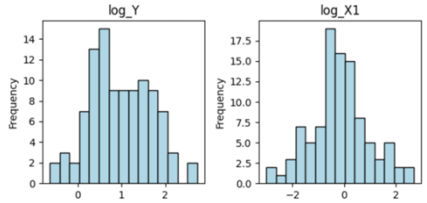
24

Visual Diagnostics

Histograms of the variables show that Y and X1 are skewed to the right while X2 is relatively symmetric.



The histogram for X2 suggests that log is not needed for X2.
Logarithms of Y and X1 are relatively symmetric. Hence, estimation with logarithms of Y and X1 seems promising.



25

Chosen Model

The visual diagnostics on the previous suggest estimating the model with Y and X1 in logged form and X2 not logged. This yields the following result:

OLS Regression Results

Dep. Variable:	log(Y)	**R-squared:**	0.863
Model:	OLS	**Adj. R-squared:**	0.860
No. Observations:	100	**F-statistic:**	364.6
Covariance Type:	HC1	**Prob (F-statistic):**	7.58e-46

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	0.9986	0.027	37.348	0.000	0.946	1.052
log(X1)	0.5164	0.023	22.526	0.000	0.471	0.562
X2	0.2866	0.024	12.057	0.000	0.239	0.334

The Ramsey test of this model yields high p-values suggesting this is a good functional form:

	power	pvalue
0	2	0.915791
1	3	0.696789

26

26

AI Prompt and Result

I issued the following prompt to AI:

To estimate a regression, I have data for three variables, Y, X1, and X2. The data are below. Find the appropriate functional form for the variables with Y as dependent variable and X1 and X2 as the independent variables.

- Below the prompt, I pasted the data.
- AI provided the following result:

Preferred functional form (econometrically sound)

Log–linear model:

$$\boxed{\ln(Y_i) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i}$$

- Let's investigate this model.

27

27

Model Chosen by AI

The model proposed by AI has the logarithm of Y with X1 and X2 not logged. This yields the following result:

OLS Regression Results

Dep. Variable:	log(Y)	**R-squared:**	0.588
Model:	OLS	**Adj. R-squared:**	0.580
No. Observations:	100	**F-statistic:**	64.91
Covariance Type:	HC1	**Prob (F-statistic):**	1.28e-18

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	0.6079	0.063	9.664	0.000	0.483	0.733
X1	0.1893	0.033	5.712	0.000	0.124	0.255
X2	0.3056	0.042	7.210	0.000	0.221	0.390

The Ramsey test of this model yields exceedingly low p-values suggesting this is not a good functional form:

	power	pvalue
0	2	0.000005
1	3	0.000004

I am surprised that AI performed so badly on such a simple application.

28

28

Building A Regression Model

In your summer internships and future work, you likely want to build regression models.

To build your model:

- Try to gain a deep understanding of the environment to guide your choice of variables.
- Gather data. This often require considerable time and effort,
- View summary statistics and histograms to get a feel for your data.
- Estimate your model. Test the functional form using the Ramsey test.
- If one or both Ramsey p-values is low, try logs of variables that are skewed to the right.
- Judge whether coefficients have algebraic signs and magnitudes that are reasonable.
- If not, think about whether there may be one or more variables you are missing.
- Use judgment about whether to retain variables that have high p-values.
- Apply your model in managerial decision making.

29

29

Summary

The objective of this slide deck has been to help you gain experience in building a regression model.

The steps we pursued in building a model for Boston Condominiums will be useful for you in building your own model.

Of course, you will have an extensive period of time to gather data and explore alternative models when you are developing your own regression model.

30

30